XIAMEN UNIVERSITY MALAYSIA

ONLINE MID-TERM TEST 1

Course Code: COM107

Course Name: Circuits and Devices

Question Paper Setter: Chua Ming Yam

Academic Session: 2022/04

Total No. of Pages: 2 Time Allocated: 1 hour

Additional Materials: All materials related to COM107.

INSTRUCTIONS TO CANDIDATES

1. This paper consists of **TWO** pages excluding the cover page.
2. There are **TWO** questions in this paper. Attempt **ALL**.
3. Show all your calculations in your answer booklet.
4. Please type in all the corresponding answers on the downloadable answer book provided.
5. Communication between candidates by any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates or suspected of doing so, you would be penalized accordingly.

(Student ID: _____ Full Name: _____)

Question 1 (50 marks, CLO1-PLO1:C3)

A coil is wound with 125 turns on a rectangle former with a length and width of 100 mm and 50 mm, respectively. The coil rotates in a uniform magnetic field of density 1.0 T. The ends of the coil are electrically connected to the terminals of a 120 Ω resistor.

a) Calculate the rotation speed of the coil in Revolution per Minute (RPM) to produce an a.c. waveform with a frequency of 60 Hz. [4 marks]

b) Calculate the period of the waveform. [4 marks]

c) Calculate the maximum value of the e.m.f. generated in the coil. [12 marks]

d) Write the instantaneous expression for the current flowing through the resistor [10 marks]

e) Determine the amount of current flows in the coil when the coil has rotated through 45° from the position of zero e.m.f. [5 marks]

f) Sketch the current waveform indicating

i. The peak amplitude of the current [3 marks]

ii. The peak-to-peak current [3 marks]

iii. The average current [3 marks]

iv. The root mean square (r.m.s.) current [4 marks]

v. The waveform period [2 marks]

Continues…

Question 2 (50 marks, CLO1-PLO1:C3)

For the following sinusoids,

$$x(t) = 230\cos\left(377t + \frac{\pi}{8}\right)$$

$$y(t) = 110\sin(377t + 80^{\circ})$$

a) Convert the sinusoids into their complex number notation. [2×10 marks]

b) Write the instantaneous expression for,

i. $a(t) = x(t) - y(t)$ [8 marks]

ii. $b(t) = x(t) \div y(t)$ [10 marks]

c) Sketch the phasor diagram for $x(t)$, $y(t)$, $a(t)$, $b(t)$, [4×3 marks]

- END OF PAPER -